205 Moment

2.5.1 · 2.5.2 · 2.5.3 Moment and perpendicular lever arm

A moment measures a force's turning effect about a pivot. Its magnitude is force times the perpendicular distance from the pivot to the force's line of action. Equivalently, use the perpendicular force component times the distance to the application point. An arbitrary sloping distance is not the lever arm. State whether the moment is clockwise or anticlockwise.

$$\tau = Fd_{\perp}$$

$$\tau = Fd\sin\theta = Fd_{\perp} = F_{\perp}d$$

τ: moment (N m); F: force (N); d⊥: shortest distance from pivot to force's line of action (m). State clockwise or anticlockwise.

τ: moment magnitude (N m); d: pivot-to-force-application distance (m); θ: angle between d and F; d⊥ = d sin θ; F⊥ = F sin θ.

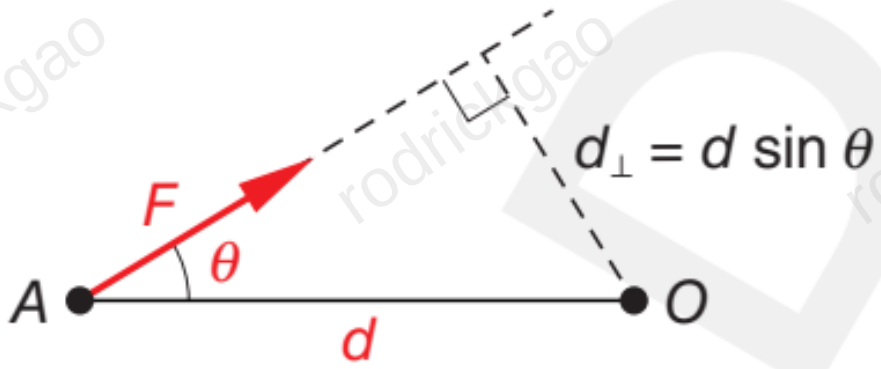


Fig 5.1c $\tau = Fd \sin\theta$.

The lever arm is the perpendicular distance from O to the force line, not simply OA. Here d⊥ = d sin θ.

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2.5.4 Conditions for rigid-body equilibrium

Equilibrium requires both zero resultant force and zero resultant moment, excluding translational and angular acceleration respectively. Resolve forces along suitable axes and choose a convenient pivot for moments. A force through the pivot contributes no moment about it, but must still appear in the force-balance equations.

$$\sum\vec{F} = \vec{0}$$

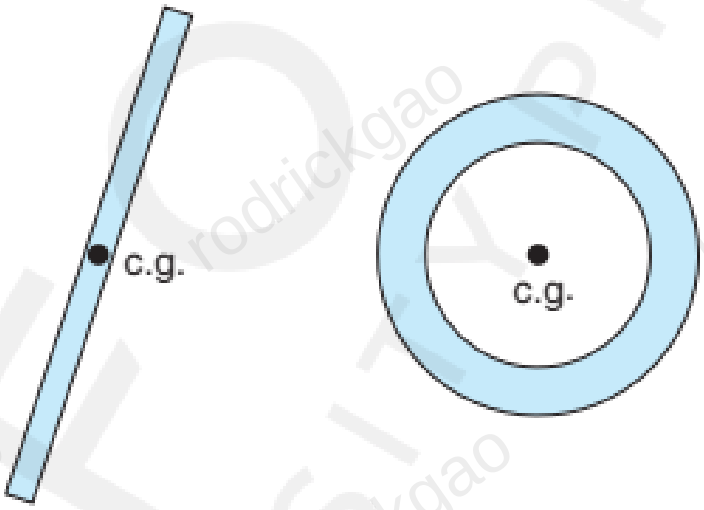
$$\sum\tau = 0$$

Equilibrium requires zero resultant force and zero resultant moment about any point. Resolve forces and use one consistent sign convention for moments.

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2.5.5 Centre of gravity and stability

The centre of gravity is the point through which the resultant weight can be treated as acting. In a uniform gravitational field, draw the weight vertically downwards from this point when calculating moments. Toppling depends on whether its vertical projection lies within the support base; a lower centre of gravity and wider base generally improve stability.



The centre of gravity of a uniform symmetrical object can lie in a space containing no material.

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